E-COMMERCE FURNITURE DATASET PROJECT

OBJECTIVE

Predict the number of furniture items sold (sold) based on product attributes such as productTitle, originalPrice, price, and tagText.

Data Collection

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("ecommerce_furniture_dataset_2024.csv")
```

```
df.head()
```

	productTitle	originalPrice	price	sold	tagText
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	NaN	$46.79	600	Free shipping
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	NaN	$169.72	0	Free shipping
2	Desser For Bedroom With 7 Fabric Drawers Organ...	$78.4	$39.46	7	Free shipping
3	Modern Accent Boucle Chair,Upholstered Tufted ...	NaN	$111.99	0	Free shipping
4	Small Unit Simple Computer Desk Household Wood...	$48.82	$21.37	1	Free shipping

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(2000, 5)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
```

```
 0   productTitle   2000 non-null   object
 1   originalPrice  487 non-null    object
 2   price          2000 non-null   object
 3   sold           2000 non-null   int64
 4   tagText        1997 non-null   object
dtypes: int64(1), object(4)
memory usage: 78.3+ KB
```

Data Preprocessing

```
df = df[['productTitle', 'originalPrice', 'price', 'tagText', 'sold']]
```

```
df.head()
```

	productTitle	originalPrice	price	tagText	sold
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	NaN	$46.79	Free shipping	600
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	NaN	$169.72	Free shipping	0
2	Desser For Bedroom With 7 Fabric Drawers Organ...	$78.4	$39.46	Free shipping	7
3	Modern Accent Boucle Chair,Upholstered Tufted ...	NaN	$111.99	Free shipping	0
4	Small Unit Simple Computer Desk Household Wood...	$48.82	$21.37	Free shipping	1

Next steps: Generate code with df New interactive sheet

```
df.isnull().sum()
```

	0
productTitle	0
originalPrice	1513
price	0
tagText	3
sold	0

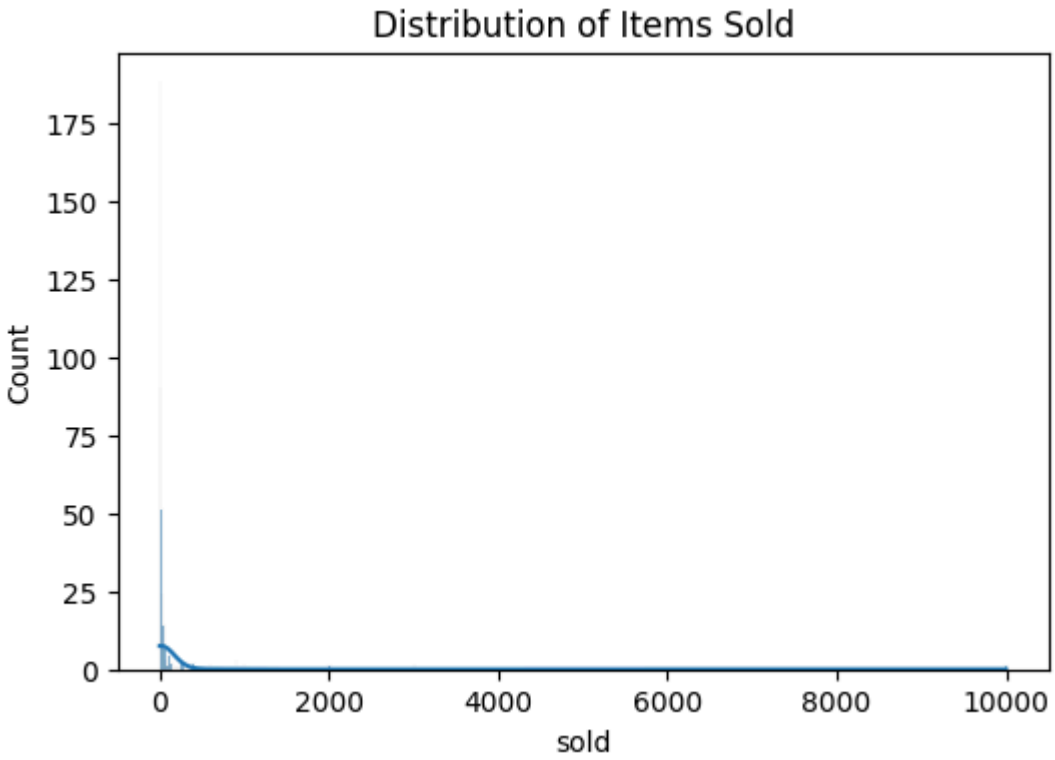
dtype: int64

```
df = df.dropna()
```

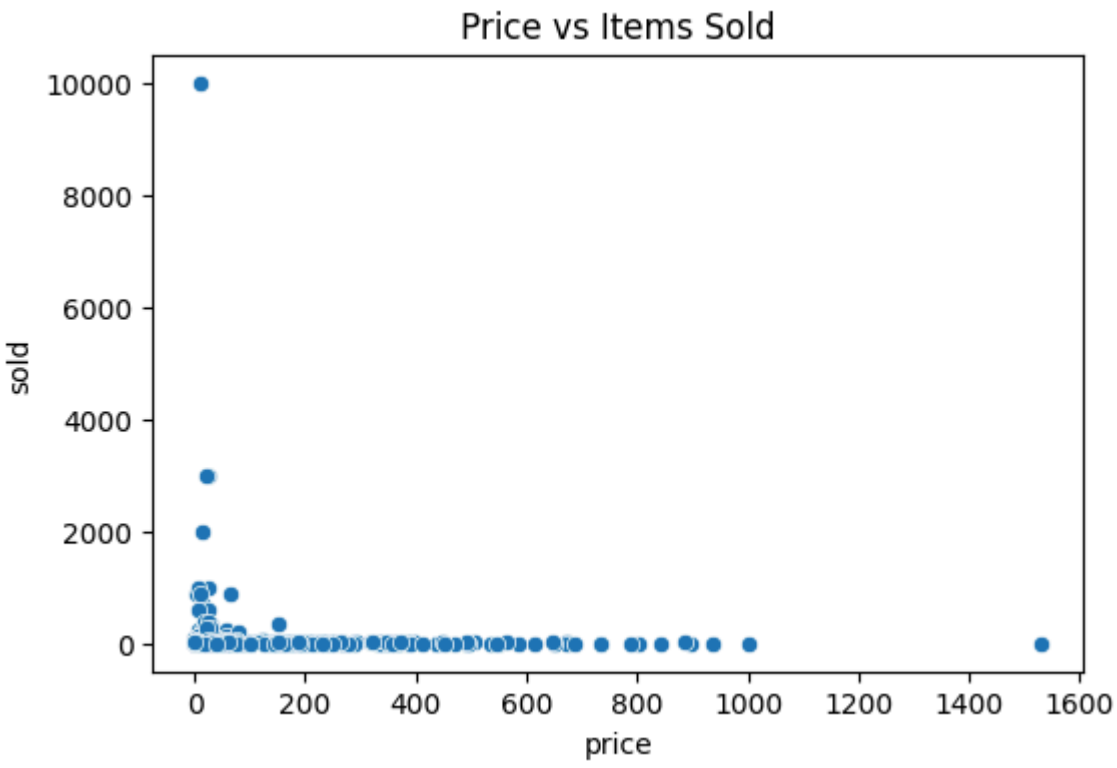
```
df['originalPrice'] = df['originalPrice'].replace('[₹, $]', '', regex=True).
df['price'] = df['price'].replace('[₹, $]', '', regex=True).astype(float)
```

Exploratory Data Analysis (EDA)

```
plt.figure(figsize=(6,4))
sns.histplot(df['sold'], kde=True)
plt.title("Distribution of Items Sold")
plt.show()
```

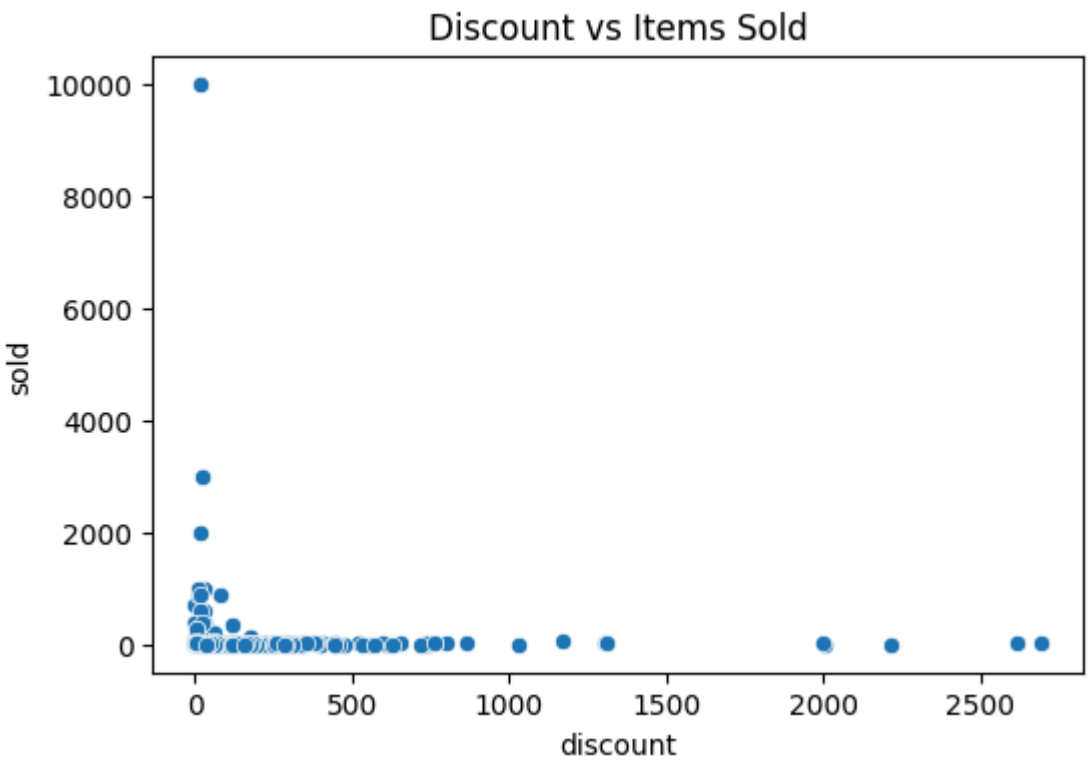



```
plt.figure(figsize=(6,4))
sns.scatterplot(x='price', y='sold', data=df)
plt.title("Price vs Items Sold")
plt.show()
```




```
df['discount'] = df['originalPrice'] - df['price']

plt.figure(figsize=(6,4))
sns.scatterplot(x='discount', y='sold', data=df)
plt.title("Discount vs Items Sold")
plt.show()
```

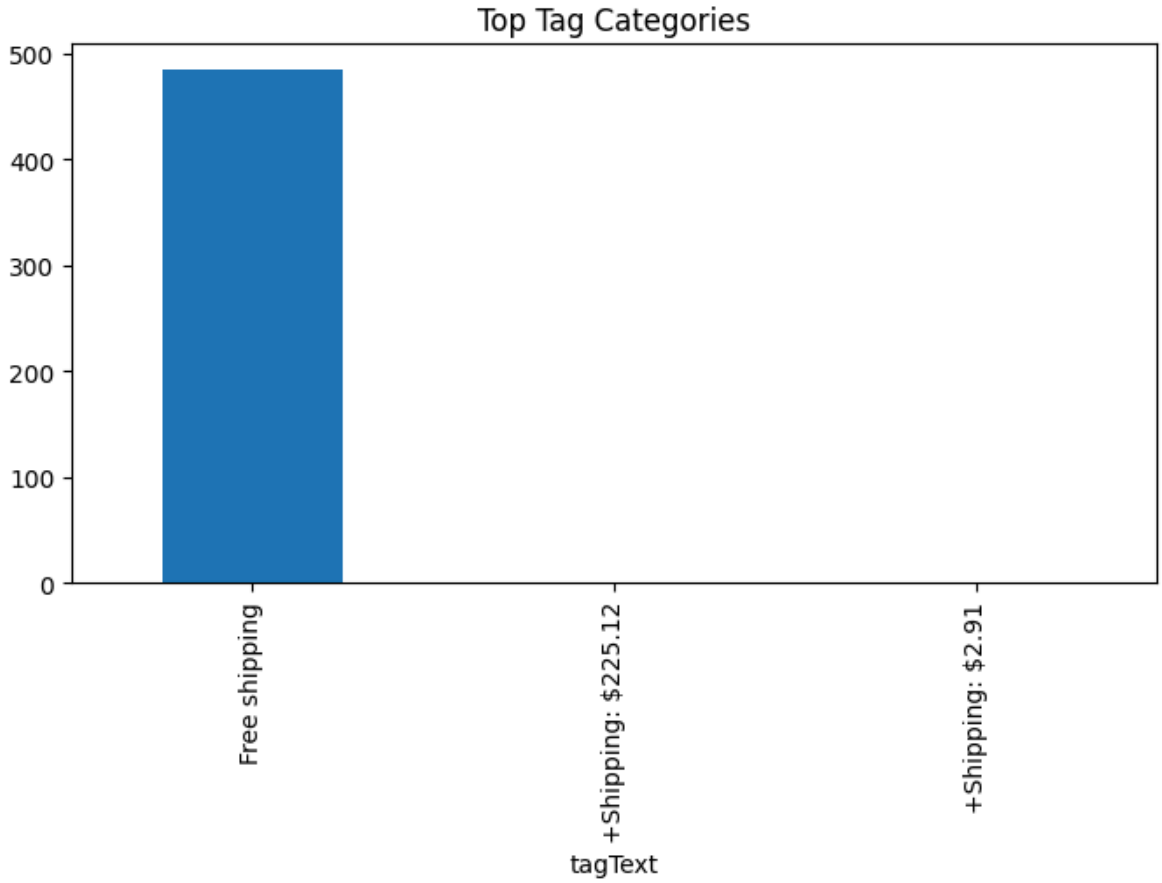


```
df['tagText'].value_counts().head(10)
```

tagText	count
Free shipping	485
+Shipping: $225.12	1
+Shipping: $2.91	1

dtype: int64

```
plt.figure(figsize=(8,4))
df['tagText'].value_counts().head(10).plot(kind='bar')
plt.title("Top Tag Categories")
plt.show()
```



Feature Engineering

```
df['discount'] = df['originalPrice'] - df['price']
df['discount_percent'] = (df['discount'] / df['originalPrice']) * 100
```

```
df_encoded = pd.get_dummies(df, columns=['tagText'], drop_first=True)
```

```
df_encoded['title_length'] = df_encoded['productTitle'].apply(len)
df_encoded = df_encoded.drop(columns=['productTitle'])
```

```
X = df_encoded.drop('sold', axis=1)
y = df_encoded['sold']
```

Model Selection & Training

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.ensemble import RandomForestRegressor

rf_model = RandomForestRegressor(n_estimators=200, random_state=42)
rf_model.fit(X_train, y_train)
```

```
▾ RandomForestRegressor
RandomForestRegressor(n_estimators=200, random_state=42)
```

Model Evaluation

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_s
import numpy as np
```

```
y_pred_rf = rf_model.predict(X_test)

print("Random Forest")
print("MAE:", mean_absolute_error(y_test, y_pred_rf))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_rf)))
print("R2 Score:", r2_score(y_test, y_pred_rf))
```

```
Random Forest
MAE: 170.58045918367344
RMSE: 595.3425632045892
R2 Score: -1.575352411474157
```

CONCLUSION

Discount and price significantly impact sales. Products with promotional tags sell more. Random Forest performs better due to nonlinear relationships.

```
Start coding or generate with AI.
```